

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q22-2260

Battlefield 138 kV
December 31, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2260. The Interconnection Customer proposes a 237.6 MW battery energy storage facility interconnecting at the Battlefield 138 kV substation in the Virginia area. The request was received on December 20, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 455 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,069,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Kewaunee to Tyler Substation 138.0 kV	North Tower Tap to McCartney 345.0 kV L	107.5	227	5.7
Deen Point to Goldfield 138.0 kV	McCartney to N Tower Tap 345.0 kV	104.0	266	5.7
Greening 345/138kV XFMR	Zephyrhills to Lamping Sta 138.0 kV	100.5	459	8.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Kewaunee to Tyler Substation 138.0 kV	Reconductor line	\$3,177,000	3% (\$83,000)
Deen Point to Goldfield 138.0 kV	Reconductor line	\$637,000	3% (\$17,000)
Greening 345/138kV TX	Replace transformer with higher-rated unit	\$21,357,000	4% (\$969,000)
	Total allocated to project		\$1,069,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.